

HOS02A Intro to Docker

AI500 – Artificial Intelligence of Cloud Computing

by Verónica Elze in Summer 2024

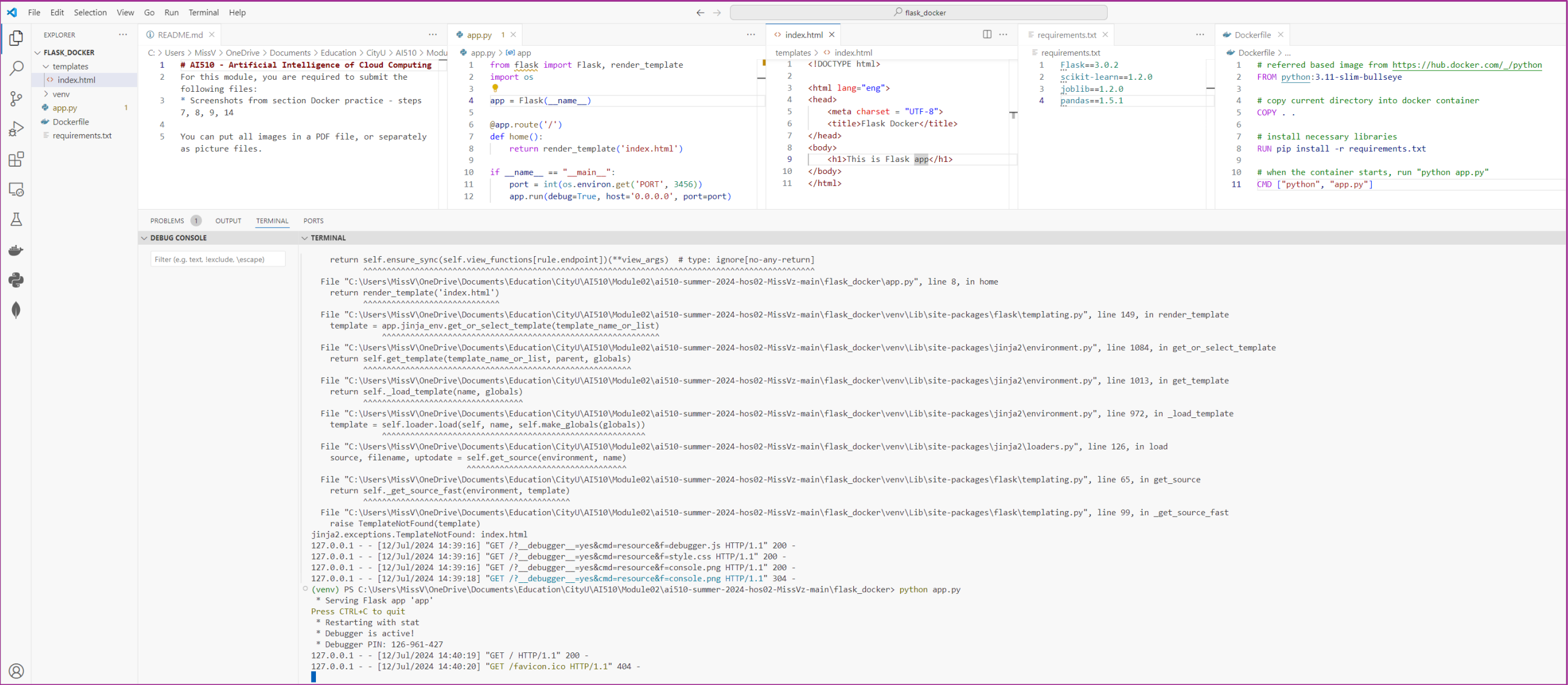


Figure 1 Step 7 Create Dockerfile

```

uvu@Desiree: ~/hos02/flask_1 × + ▾
uvu@Desiree:~/hos02$ ls
flask_docker
uvu@Desiree:~/hos02$ cd flask_docker
uvu@Desiree:~/hos02/flask_docker$ ls
Dockerfile app.py requirements.txt templates venv
uvu@Desiree:~/hos02/flask_docker$ sudo docker image build -t flask_docker_image .
[+] Building 46.0s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 338B                             0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim-bullseye 2.3s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [internal] load build context                                0.3s
=> => transferring context: 28.67MB                             0.3s
=> [1/3] FROM docker.io/library/python:3.11-slim-bullseye@sha256:7c0ce75dbdab24e15a1a7e5e89ee2e93833ec935348643c 7.0s
=> => resolve docker.io/library/python:3.11-slim-bullseye@sha256:7c0ce75dbdab24e15a1a7e5e89ee2e93833ec935348643c 0.0s
=> => sha256:7dabe778191072ed3f217b15454db0897c0a53b95ce621f4aed2185d2dec725e 12.02MB / 12.02MB 3.8s
=> => sha256:7c0ce75dbdab24e15a1a7e5e89ee2e93833ec935348643c34455a06bec249865 9.12kB / 9.12kB 0.0s
=> => sha256:cd0f7f666061e0009d315d1d7b9bf7a2a393f9c9573bee106b6aaa2947a416e9 1.94kB / 1.94kB 0.0s
=> => sha256:d1510951db9e079f7329c1435e6febbfcee8275e9dff576cbae4431e112ea74f 6.92kB / 6.92kB 0.0s
=> => sha256:76956b537f14770ffd78afbe4f17016b2794c4b9b568325e8079089ea5c4e8cd 31.42MB / 31.42MB 4.3s
=> => sha256:8c100e54bf68eb70353bc6b5a7af71c0eab4eacc38b0085892096c2de7ce747f 1.08MB / 1.08MB 1.3s
=> => sha256:5aaf9b2292424d13c80685eee26a3418f7a6c4817849ab855fc201616c23fba2 230B / 230B 1.6s
=> => sha256:5104a32e73cb7b739f7b53a47abd37fd67581bf2665c7cdfb743fb5f1d80070d 3.21MB / 3.21MB 3.0s
=> => extracting sha256:76956b537f14770ffd78afbe4f17016b2794c4b9b568325e8079089ea5c4e8cd 1.5s
=> => extracting sha256:8c100e54bf68eb70353bc6b5a7af71c0eab4eacc38b0085892096c2de7ce747f 0.1s
=> => extracting sha256:7dabe778191072ed3f217b15454db0897c0a53b95ce621f4aed2185d2dec725e 0.5s
=> => extracting sha256:5aaf9b2292424d13c80685eee26a3418f7a6c4817849ab855fc201616c23fba2 0.0s
=> => extracting sha256:5104a32e73cb7b739f7b53a47abd37fd67581bf2665c7cdfb743fb5f1d80070d 0.3s
=> [2/3] COPY . . 0.4s
=> [3/3] RUN pip install -r requirements.txt 34.5s
=> exporting to image 1.7s
=> => exporting layers 1.7s
=> => writing image sha256:5ffebdf37d7bf0c317d11934935b7203912c3a565cf99bad56d48fc96debbf6b 0.0s
=> => naming to docker.io/library/flask_docker_image 0.0s
uvu@Desiree:~/hos02/flask_docker$ |

```

Figure 2 Step 8 Build Docker image


```
uvu@Desiree: ~/hos02/flask_1 X + v
=> => transferring dockerfile: 338B 0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim-bullseye 2.3s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [internal] load build context 0.3s
=> => transferring context: 28.67MB 0.3s
=> [1/3] FROM docker.io/library/python:3.11-slim-bullseye@sha256:7c0ce75dbdab24e15a1a7e5e89ee2e93833ec935348643c 7.0s
=> => resolve docker.io/library/python:3.11-slim-bullseye@sha256:7c0ce75dbdab24e15a1a7e5e89ee2e93833ec935348643c 0.0s
=> => sha256:7dabe778191072ed3f217b15454db0897c0a53b95ce621f4aed2185d2dec725e 12.02MB / 12.02MB 3.8s
=> => sha256:7c0ce75dbdab24e15a1a7e5e89ee2e93833ec935348643c34455a06bec249865 9.12kB / 9.12kB 0.0s
=> => sha256:cd0f7f666061e0009d315d1d7b9bf7a2a393f9c9573bee106b6aaa2947a416e9 1.94kB / 1.94kB 0.0s
=> => sha256:d1510951db9e079f7329c1435e6febbfcee8275e9dff576cbae4431e112ea74f 6.92kB / 6.92kB 0.0s
=> => sha256:76956b537f14770ffd78afbe4f17016b2794c4b9b568325e8079089ea5c4e8cd 31.42MB / 31.42MB 4.3s
=> => sha256:8c100e54bf68eb70353bc6b5a7af71c0eab4eacc38b0085892096c2de7ce747f 1.08MB / 1.08MB 1.3s
=> => sha256:5aaf9b2292424d13c80685eee26a3418f7a6c4817849ab855fc201616c23fba2 230B / 230B 1.6s
=> => sha256:5104a32e73cb7b739f7b53a47abd37fd67581bf2665c7cdfb743fb5f1d80070d 3.21MB / 3.21MB 3.0s
=> => extracting sha256:76956b537f14770ffd78afbe4f17016b2794c4b9b568325e8079089ea5c4e8cd 1.5s
=> => extracting sha256:8c100e54bf68eb70353bc6b5a7af71c0eab4eacc38b0085892096c2de7ce747f 0.1s
=> => extracting sha256:7dabe778191072ed3f217b15454db0897c0a53b95ce621f4aed2185d2dec725e 0.5s
=> => extracting sha256:5aaf9b2292424d13c80685eee26a3418f7a6c4817849ab855fc201616c23fba2 0.0s
=> => extracting sha256:5104a32e73cb7b739f7b53a47abd37fd67581bf2665c7cdfb743fb5f1d80070d 0.3s
=> [2/3] COPY . . 0.4s
=> [3/3] RUN pip install -r requirements.txt 34.5s
=> exporting to image 1.7s
=> => exporting layers 1.7s
=> => writing image sha256:5ffebdf37d7bf0c317d11934935b7203912c3a565cf99bad56d48fc96debbf6b 0.0s
=> => naming to docker.io/library/flask_docker_image 0.0s
uvu@Desiree:~/hos02/flask_docker$ sudo docker run --rm -p 3456:3456 flask_docker_image
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:3456
* Running on http://172.17.0.3:3456
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 756-558-930
172.17.0.1 - - [12/Jul/2024 23:46:24] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [12/Jul/2024 23:46:25] "GET /favicon.ico HTTP/1.1" 404 -
```

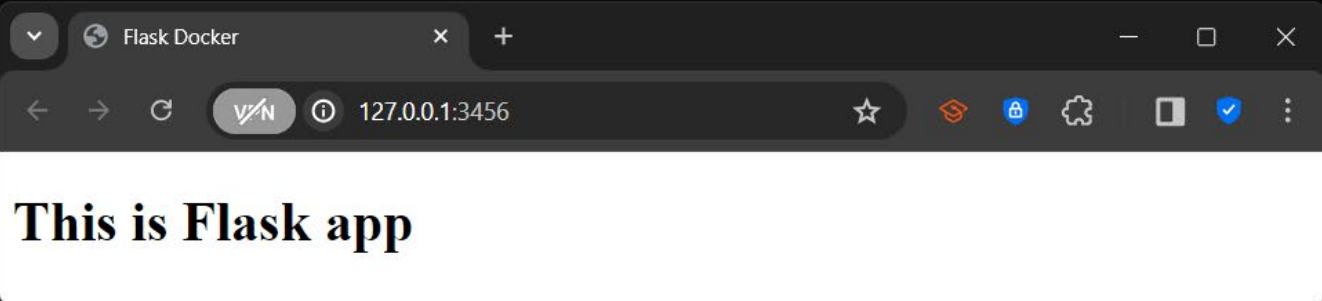


Figure 3 Step 9 Start Docker image

```
✓ TERMINAL bash - uvu + v [ ] [ ]

1.6s
=> => writing image sha256:4e99a5d6f7af4e113466f5e985f1aecf26958a1fbd3ad9a8bf40179eb5191b58
0.0s
=> => naming to docker.io/library/flask_docker_image
0.0s
• uvu@Desiree:~/hos02/flask_docker$ sudo docker run --rm -p 3456:3456 flask_docker_image

/usr/local/lib/python3.11/site-packages/sklearn/base.py:288: UserWarning: Trying to unpickle estimator SVC from version 1.5.0 when using version 1.2.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:3456
* Running on http://172.17.0.3:3456
Press CTRL+C to quit
* Restarting with stat
/usr/local/lib/python3.11/site-packages/sklearn/base.py:288: UserWarning: Trying to unpickle estimator SVC from version 1.5.0 when using version 1.2.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
* Debugger is active!
* Debugger PIN: 125-940-236
172.17.0.1 - - [13/Jul/2024 01:22:30] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [13/Jul/2024 01:22:31] "GET /favicon.ico HTTP/1.1" 404 -
172.17.0.1 - - [13/Jul/2024 01:23:43] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [13/Jul/2024 01:24:19] "GET /predict/10 HTTP/1.1" 200 -
```

   127.0.0.1:3456/predict/10 

Prediction result: [1]

Figure 4 Step 14 Test predict path